

September 2017 examination

EC400

Probability and Statistical Inference

Suitable for all candidates

Instructions to candidates

This paper contains **SEVEN** questions. Answer **ALL** questions. Write your name, your class group number, and the name of your class teacher on your script.

Time Allowed	Reading Time:	<i>None</i>
	Writing Time:	<i>2 hours</i>

You are supplied with:	<i>Statistical Tables</i>
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You may also use:	<i>No additional materials</i>
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Calculators:	<i>Calculators are not allowed in this examination</i>
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1. **(7 points - 10 minutes)** Suppose that 1 in 10,000 people (0.01%) has a particular disease. If a person has the disease, the test will give a positive result with a probability of 0.99. If a person does not have the disease, the test will give a negative result with a probability of 0.99. Find the probability of not having the disease if the test comes out positive. Provide your answer as a simplified fraction.
2. **(8 points - 10 minutes)** Let X be a continuous random variable with pdf $f_X(x) = \frac{c}{x^2}$ if $1 < x < \infty$ and $f_X(x) = 0$ otherwise.

- (a) (2 points) Determine the value of the constant c .
- (b) (6 points) Consider the continuous random variable $Y = \exp(\sqrt{X})$. What is the pdf of Y and what is the probability of $Y > 0$?

3. **(18 points - 20 minutes)** Let X have an exponential distribution with parameter $\lambda > 0$. Its density is given by

$$f(x) = \frac{1}{\lambda} \exp\left(-\frac{1}{\lambda}x\right), 0 \leq x < \infty, 0 \text{ otherwise,}$$

and $E(X) = \lambda$. Let $X_1, \dots, X_n$ be an iid sample drawn from this distribution.

- (a) (8 points) Derive the maximum likelihood estimator for λ ($\hat{\lambda}_{MLE}$).
 - (b) (5 points) The variance of X is given by λ^2 . Discuss how you can obtain a consistent estimator for the variance. Explain your claim clearly.
 - (c) (5 points) Provide a method of moment estimator for the variance of X . Explain the intuition underlying your proposal.
4. **(12 points - 15 minutes)** Let $f(x, y) = 3/2, x^2 \leq y < 1, 0 \leq x \leq 1$, be the joint pdf of X and Y .
 - (a) (5 points) Find the conditional distribution of Y given $X = x$.
 - (b) (2 points) Are X and Y independent.
 - (c) (5 points) Find $\Pr(X \geq 1/2, Y \geq 1/2)$. You may want to use a sketch when answering this question.

5. **(25 points - 30 minutes)** Let us assume that the continuous random variable X "Height of an adult living in the UK" has a normal distribution with mean μ_X and variance $\sigma_X^2 = 0.025$. Say a random sample of 1000 observations of heights of people in the UK was obtained, and the sample mean was given by $\bar{x} = 1.769$.
- (a) (6 points) Test the hypothesis that $H_0 : \mu_X = 1.77$ against the alternative $H_A : \mu_X \neq 1.77$ using a significance level of 5%.
 - (b) (6 points) Explain how to compute the p-value for the two-sided test. Draw a sketch to illustrate. **(There is no need to provide an exact numerical answer here or in the rest of the question).**
 - (c) (7 points) Discuss how you would obtain the power of your test in (a) if the true mean adults' height is 1.765.
 - (d) (6 points) Say, in (a) we had tested the hypothesis that $H_0 : \mu_X = 1.77$ against the alternative $H_A : \mu_X > 1.77$. What can you then say about power of your test if the true mean adults' height is 1.765? Discuss this finding (compare it to the significance level). You may want to draw a sketch to illustrate.
6. **(18 points - 20 minutes)** Let $X_1, \dots, X_n$ an iid sample drawn from the distribution $X \sim N(0, \sigma^2)$ where σ^2 is unknown but finite.
- (a) (10 points) Drawing on the standard estimator for the sample variance, someone proposes to use the following estimator for σ^2 :

$$\tilde{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n X_i^2.$$

Is this estimator unbiased and consistent? Proof your claims. Discuss the difference between these two concepts.

- (b) (8 points) Instead of providing an estimate of σ^2 for a given sample, let $n = 10$, you are asked to construct a 95% confidence interval. You may continue assuming that μ is known and equal to 0. What details do you need to compute the confidence interval and what information does the interval provide?
Hint: You may recall $\frac{1}{\sigma^2} \sum_{i=1}^n X_i^2 \sim \chi_n^2$.

7. **(12 points - 15 minutes)** Let $f(x)$ and $F(x)$ denote, respectively, the pdf and the cdf of the random variable X .

(a) (4 points) Define the conditional pdf of X given $X > x_0$ where x_0 a fixed number. This function, also called the Hazard function, finds application in a problem of the time until death, given survival until time x_0 and show that $f(x|X > x_0)$ is a pdf.

(b) (8 points) Let $f(x) = e^{-x}$, $0 < x < \infty$, zero elsewhere. Compute $\Pr(X > 2|X > 1)$. (You are not expected to provide an exact numerical answer).

TABLE FD.1 Observations on Income and Education, 20 Observations

$I =$ Observation,

$$Y = \text{Income.}$$

Source: Data are artificial.

APPENDIX G



STATISTICAL TABLES

TABLE G.1 Cumulative Normal Distribution. Table Entry Is $\Phi(z) = \text{Prob}[Z \leq z]$

[illegible]